

Computer Ports

Lesson 16 — Physical Connectors and Logical Ports

JKSSB Computer Awareness

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What Is a Port?

- ▶ A **port** is a socket or interface on a device where a cable or another device is connected.
- ▶ A port lets data and/or power move between the computer and the attached device.
- ▶ Ports differ by **shape**, by signal type (digital or analog), and by purpose.
- ▶ Common physical ports on a PC: USB, HDMI, DisplayPort, VGA, Ethernet (RJ-45), audio jack, and SATA.
- ▶ Keep the words apart: the **port** is the socket, the **connector** is the plug end of the cable.

The one idea

A port is the doorway between the computer and a device; the connector is the plug that fits that doorway.

The Two Meanings of “Port”

- ▶ **Physical port**: a hardware socket, such as a USB port, an HDMI port, or an RJ-45 port.
- ▶ **Logical port** (network port): a software number that identifies a service running on a computer — it is not a socket.
- ▶ The same machine has both: web traffic uses logical port 80, while the network cable still plugs into a physical RJ-45 port.

Trap alert

In networking, “port” means a **number** (0–65535), not a connector. Do not confuse a port number with a port on the back of the computer.

USB — Universal Serial Bus

- ▶ **USB** stands for Universal Serial Bus.
- ▶ It is a standard interface for connecting many peripherals to a computer.
- ▶ It supports **plug and play** and **hot swapping** — a device can be attached or removed without switching the computer off.
- ▶ A USB connection carries both **data** and **power**.
- ▶ Typical devices: keyboard, mouse, printer, pen drive, external disk, webcam, and smartphone.

USB Connector Types

- ▶ **USB Type-A**: the familiar rectangular connector on most PCs.
- ▶ **USB Type-B**: often used for printers and scanners.
- ▶ **Mini-USB** and **Micro-USB**: smaller connectors used on older phones and devices.
- ▶ **USB Type-C**: a modern, small, reversible connector.

The connector shape can change while the underlying USB standard stays the same.
Do not treat a connector shape as the bus itself.

USB-C (USB Type-C)

- ▶ **USB-C** is a connector type, not a separate bus.
- ▶ It is **reversible**: it can be plugged in either way up.
- ▶ One connector can carry **data**, **power**, and **video**.
- ▶ Typical devices: modern smartphones, laptops, external SSDs, and docking stations.

Keep it straight

USB is the standard; USB-C is one connector shape for that standard.

HDMI

- ▶ **HDMI** stands for High-Definition Multimedia Interface.
- ▶ It carries **digital video and digital audio** over one cable.
- ▶ Typical devices: monitor, television, projector, set-top box, and game console.
- ▶ A single HDMI cable replaces separate video and audio cables.

In the office

The desktop is linked to a large screen with one HDMI cable, so both the picture and the sound travel over the same wire.

DisplayPort

- ▶ **DisplayPort** is a **digital** audio/video interface designed mainly for computers and monitors.
- ▶ It is common on graphics cards, laptops, and desktop monitors.
- ▶ It supports high refresh rates and multiple monitors through daisy-chaining.
- ▶ **Mini DisplayPort** is a smaller variant of the same interface.

HDMI vs DisplayPort

Point	HDMI	DisplayPort
Signal	Digital video + audio	Digital video + audio
Main use	TV, home theatre, monitor	PC monitors, graphics cards
Note	Very common on TVs	Common on PCs; multi-monitor

Both are digital. The right choice usually depends on the device, not on which one is somehow “better”.

VGA

- ▶ **VGA** stands for Video Graphics Array.
- ▶ It is an **analog** video interface — video only, no audio.
- ▶ It uses a 15-pin D-sub connector, often blue, held by two screws.
- ▶ Found on older monitors and projectors; now largely replaced by HDMI and DisplayPort.

Trap alert

VGA is **analog** and carries **no audio**. HDMI and DisplayPort are digital and do carry audio.

Ethernet and RJ-45

- ▶ **Ethernet** is the standard for **wired** computer networking (IEEE 802.3).
- ▶ **RJ-45** (Registered Jack 45) is the 8-pin connector used with twisted-pair LAN cable.
- ▶ Typical devices: router, switch, modem, and the desktop network card.

Trap alert

RJ-45 (network, 8 pins) is **not** RJ-11 (the smaller telephone jack). The two are near-neighbours in shape but different in use.

Audio Jacks

- ▶ The common PC **3.5 mm audio jack** connects headphones, earphones, speakers, and microphones.
- ▶ **TRS** (Tip-Ring-Sleeve) carries stereo audio.
- ▶ **TRRS** (Tip-Ring-Ring-Sleeve) adds a microphone channel.
- ▶ Colour convention on PCs: green = audio out, pink = microphone.

Some equipment uses 2.5 mm or 6.35 mm jacks instead of 3.5 mm.

SATA and eSATA

- ▶ **SATA** stands for Serial Advanced Technology Attachment.
- ▶ It connects **internal storage** — a hard disk, an SSD, or an optical drive — to the motherboard.
- ▶ The SATA data cable is thin and has 7 pins; a separate power connector feeds the drive.
- ▶ **eSATA** (external SATA) is used for an external drive; it is a **storage interface**, not a general peripheral port like USB.

Match the Port to Its Use

Port	Typical use
USB / USB-C	Keyboard, pen drive, phone; data + power
HDMI	Digital audio + video to a TV or monitor
DisplayPort	Digital audio + video to PC monitors
VGA	Analog video to older monitors and projectors
RJ-45 (Ethernet)	Wired network (LAN) cable
3.5 mm jack	Headphones, speakers, microphone
SATA	Internal HDD, SSD, or optical drive

Term Precision: Port, Connector, Cable

- ▶ **Port**: the socket on the device.
- ▶ **Connector**: the plug end of a cable (for example, a USB-C connector).
- ▶ **Cable**: the wire joining the two.
- ▶ **USB** is a standard; **USB-C** is a connector type; a **USB port** is the socket.

Same discipline as file *format* vs file *extension* (TRM-09): always say exactly which one you mean.

Trap Alert: Physical vs Logical Ports

- ▶ A **physical port** is a hardware socket — USB, HDMI, RJ-45.
- ▶ A **logical (network) port** is a number that names a service.
- ▶ Well-known examples: HTTP 80, HTTPS 443, FTP 21, SMTP 25, DNS 53.

Trap alert

“Port 80” is a number, not a socket. A USB port or an RJ-45 port is hardware. Never mix the two meanings of the word “port”.

Lesson 16: Recall

- ▶ A port is a socket for connecting devices; a connector is the plug end of a cable.
- ▶ USB = Universal Serial Bus: plug and play, hot swapping, data + power; USB-C is reversible.
- ▶ HDMI and DisplayPort carry digital audio and video; VGA is analog video only.
- ▶ RJ-45 (Ethernet) is the wired-network connector; RJ-11 is the smaller telephone jack.
- ▶ 3.5 mm jack: TRS for stereo, TRRS adds a microphone.
- ▶ SATA connects internal storage; eSATA is for external storage.
- ▶ Physical port = a hardware socket; logical port = a network service number.